

Question 23 (a)

Criteria	Marks
• Provides correct solution	2
• Finds the depth of water at either high or low tide, or equivalent merit	1

Sample answer:

$$d = 1.3 - 0.6 \cos\left(\frac{4\pi}{25}t\right)$$

Maximum depth (*high tide*) = $1.3 + 0.6 = 1.9$ m

Minimum depth (*low tide*) = $1.3 - 0.6 = 0.7$ m

Question 23 (b)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$\begin{aligned}\text{Period} &= \frac{2\pi}{\left(\frac{4\pi}{25}\right)} \text{ hours} \\ &= 12.5 \text{ hours}\end{aligned}$$

Question 23 (c)

Criteria	Marks
• Provides correct solution	3
• Finds one value of t for $d = 1$, or equivalent merit	2
• Attempts to solve the given equation when $d = 1$	1

Sample answer:

$$d = 1.3 - 0.6 \cos\left(\frac{4\pi}{25}t\right)$$

$$1 = 1.3 - 0.6 \cos\left(\frac{4\pi}{25}t\right), \quad \text{when } d = 1$$

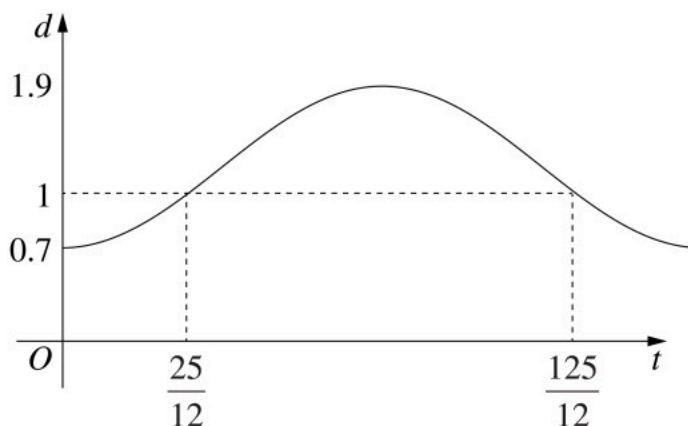
$$-0.3 = -0.6 \cos\left(\frac{4\pi}{25}t\right)$$

$$\frac{1}{2} = \cos\left(\frac{4\pi}{25}t\right)$$

$$\frac{4\pi}{25}t = \frac{\pi}{3}, \frac{5\pi}{3}$$

$$t = \frac{\pi}{3} \times \frac{25}{4\pi}, \frac{5\pi}{3} \times \frac{25}{4\pi}$$

$$= \frac{25}{12}, \frac{125}{12}$$



$$\begin{aligned} \text{Time interval} &= \frac{125}{12} - \frac{25}{12} \\ &= 8\frac{1}{3} \text{ hours} \end{aligned}$$